

Every two-vertex premaniplex is the symmetry type graph of a finite abstract polytope

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Abstract

For every rank $r \geq 3$, there are $2^r - 1$ connected two-vertex premaniplexes, one for each nonempty set of colors assigned to links between the two vertices. Pellicer, Potočník and Toledo proved that every such premaniplex occurs as the symmetry type graph of a finite maniplex. The corresponding realization problem for finite abstract polytopes remained open. Mochán established polytopal realizations when the link-color set has size one or two, or is an interval, and reduced the remaining cases in his voltage construction to one family of path-intersection identities.

We prove that every connected two-vertex premaniplex is the full symmetry type graph of a finite abstract polytope. The main step is a support-localization argument for the recursive regular polytopes used by Mochán. It treats an arbitrary set of internal link colors simultaneously and establishes the missing $k = 1$ path-intersection identity. The two exceptional rank-four base types are handled by exact finite computation. Thus the two-vertex realization problem has an affirmative answer in every rank at least three.

1 Introduction

An abstract polytope is a ranked, strongly flag-connected partially ordered set satisfying the diamond condition. Its flags form a properly edge-colored graph: two flags are joined by an edge of color i when they differ only in their rank- i face. The quotient of this flag graph by the full automorphism group is the symmetry type graph of the polytope. Its vertices are the flag orbits, and a color is represented either by a link between two orbits or by a semi-edge at one orbit.

Premaniplexes axiomatize the colored quotient graphs that may occur in this way. For a rank- r premaniplex with exactly two vertices, every color $i \in \{0, \dots, r-1\}$ is either a link between the two vertices or a semi-edge at both vertices. The graph is connected precisely when at least one color is a link. Hence there are $2^r - 1$ connected two-vertex candidates.

Pellicer, Potočník and Toledo [4] proved that all these candidates are realized by finite maniplexes. Polytopality is stronger: a maniplex is the flag graph of an abstract polytope exactly when it satisfies the path-intersection property of Garza-Vargas and Hubard [1]. Hubard and Mochán [2] translated this property into an intersection condition for voltage sets over a premaniplex.

Mochán [3] applied that framework to the two-vertex problem. He constructed finite polytopes when the link-color set has one or two elements or is an interval. For a general two-vertex type whose extreme link colors have been normalized to 0 and n , his Theorem 6.4 proves every required voltage intersection with $k > 1$. The unresolved part is the $k = 1$ identity. The difficulty noted in [3] is that an internal link of color p has voltage involving $r_0 r_p$, so a path that avoids color 0 may nevertheless move 0-faces.

The present argument controls exactly that movement. After deleting color 0, the upper voltage group is generated by local semi-edge voltages and one fundamental-cycle voltage for every internal link.

All these generators preserve a pair of endpoint regions in the recursive $\widehat{2}$ construction. Commutation with regular automorphisms then converts this coarse endpoint control into the exact path-intersection identity.

Our main result is the following.

Theorem 1. *For every $r \geq 3$, every connected rank- r two-vertex premaniplex is the symmetry type graph, with respect to the full automorphism group, of a finite abstract polytope.*

The proof in ranks at least five is structural. Two rank-four types not covered by the previously known families are finite computer-assisted base cases. The computation is described in Section 7, including its reproducibility and independence limitations.

2 Premaniplexes, voltages and the intersection criterion

We use right actions on flags. If M is a regular rank- r polytope, write

$$\text{Mon}(M) = \langle r_0, \dots, r_{r-1} \rangle$$

for its monodromy group, where r_i sends a flag to its i -adjacent flag. Write ρ_i for the corresponding distinguished automorphisms. Monodromies commute with automorphisms on the flag set.

Let X be a premaniplex and let $\Pi_I^{s,t}(X)$ denote the set of maniplex-homotopy classes of paths from the vertex s to the vertex t using only colors in I . A voltage assignment is an antimorphism

$$\xi : \Pi(X) \longrightarrow G.$$

Thus, if paths are traversed from left to right, their dart voltages multiply in reverse order. The derived colored graph X^ξ has vertices (s, g) and lifts each dart according to its voltage.

For the voltage assignments used below, the local simplicity and distant-color square conditions are already part of Mochán's construction. The remaining polytopality criterion is the following interval form of the weak path-intersection property [2, Theorem 4.2]: for all vertices s, t and all $0 \leq k, m \leq n$,

$$\xi(\Pi_{[0,m]}^{s,t}(X)) \cap \xi(\Pi_{[k,n]}^{s,t}(X)) = \xi(\Pi_{[k,m]}^{s,t}(X)). \quad (1)$$

Here $[k, m]$ is empty when $k > m$; the voltage set of the empty color set is the identity when $s = t$ and is empty otherwise.

For a two-vertex premaniplex X , denote its two vertices by a and b , and denote the set of link colors by $L(X)$. Every other color is a semi-edge at both vertices. We shall first solve the normalized core

$$L(X) = \{0, n\} \cup P, \quad P \subseteq \{1, \dots, n-1\}, \quad (2)$$

where $n \geq 4$. Lower-rank cores and colors outside the extreme links are treated in Section 8.

3 The recursive regular-polytope construction

We recall only the parts of Mochán's construction [3, Sections 4–6] that enter the new argument. Starting from a square, recursively apply the $\widehat{2}$ operation to obtain finite regular polytopes M_n . A flag of $M_n = \widehat{2}M_{n-1}$ can be written as

$$(\phi, x), \quad x \in \mathbb{F}_2^{\text{Fac}(M_{n-1})}.$$

The support $\text{supp}(x)$ is the set of facets of M_{n-1} on which x is nonzero. The construction is a lattice, and equality of lower faces is governed by containment of this support in facet stars. Mochán defines a distinguished involutory flag-permutation $T = \tilde{\rho}_0$. Its restriction to each facet is the facet automorphism reflecting across the chosen base edge; T need not be a global automorphism of M_n . Corollary 5.2 gives the local feature used below: whenever the support is contained in the union of the facet stars of two faces, the current facet has the original base edge and T acts there as the prescribed reflection.

Let X_n be obtained from X by deleting its color- n link. Mochán's construction starts with a covering

$$p : M_n \longrightarrow X_n.$$

Color the flags of M_n white or black according as their images under p are a or b , and choose the base flag Φ to be white. Every monodromy of M_n either preserves both color classes or interchanges them. The voltages below preserve the two classes. In particular, projecting a flag path in M_n through p records both its color word and whether its endpoints project to the same or different vertices of X .

Let $\Phi = (\phi, 0)$ be the base flag of M_n . Put

$$q_a = 1, \quad q_b = r_0, \quad y = Tr_0. \quad (3)$$

The involutions T and r_0 commute, so the order in the last product is immaterial. The color- n link has voltage y . For every dart d of color $c < n$, directed from s to t , Mochán's voltage assignment satisfies

$$\xi(d) = q_t r_c q_s. \quad (4)$$

This includes the color-0 link, whose two directed voltages are both trivial. If a path W from s to t has color word $c_1 \cdots c_\ell$ and

$$h = r_{c_1} \cdots r_{c_\ell},$$

then telescoping (4), with the antimorphism convention, gives

$$\xi(W) = q_t h^{-1} q_s. \quad (5)$$

Mochán also augments ξ by a finite cyclic coordinate to obtain ξ' . This auxiliary coordinate prevents an unwanted automorphism from interchanging the two fibers, so that the full symmetry type graph of the derived maniplex is exactly X [3, equation (3) and Corollary 6.5]; [4, Propositions 19 and 20 and Theorems 29 and 31]. For $m < n$, paths using colors in $[0, m]$ do not change the auxiliary coordinate. Consequently, once (1) is proved for ξ , the projection argument of [3, Corollary 6.5] proves it for ξ' as well.

Finally, [3, Theorem 6.4] proves (1) for every $k > 1$, every m , every pair of endpoints and every link set of the form (2). Thus only $k = 1$ remains.

4 The upper voltage group for arbitrary internal links

Delete color 0 from X and choose the color- n link as a spanning tree. Let

$$H_P = \xi(\Pi_{[1,n]}^{a,a}(X))$$

be the closed upper voltage group at a .

Lemma 2. *The group H_P is generated by*

$$\begin{aligned} r_i, Tr_iT & \quad (i \in \{1, \dots, n-1\} \setminus P), \\ Tr_p & \quad (p \in P). \end{aligned} \tag{6}$$

Proof. A semi-edge of color i at a contributes r_i . The corresponding semi-edge at b , transported back along the spanning-tree link, contributes Tr_iT . For an internal link of color p , the fundamental circuit formed with the spanning-tree link has voltage

$$y(r_0r_p) = Tr_p.$$

A connected two-vertex graph with a fixed spanning tree has fundamental group generated by all its semi-edges and by one fundamental circuit for every non-tree link. Therefore the displayed list is complete. In particular, multiple internal links introduce no additional mixed generator. $\square$

The importance of Lemma 2 is that every generator admits the same local support control, regardless of the size or shape of P .

5 Endpoint-support localization

Let e be the base edge of M_{n-1} , with endpoints u and v . Let the base two-face be the square with cyclically ordered vertices

$$u, v, w, q.$$

For a vertex z of M_{n-1} , write $\mathcal{F}(z)$ for the set of facets incident with z .

Lemma 3 (localization). *For every $h \in H_P$:*

1. *the first-coordinate vertices of Φh and $\Phi^1 h$ lie in $\{u, v\}$;*
2. *the first-coordinate vertex of $\Phi^{010} h$ lies in $\{w, q\}$;*
3. *in each case the vector-coordinate difference is supported in the union of the facet stars of the indicated pair.*

Proof. We induct on a word in the generators (6) and their inverses. For $i < n-1$, the monodromy r_i does not change the first-coordinate vertex or enlarge its support outside the current endpoint pair. The monodromy r_{n-1} changes one vector coordinate belonging to a facet through the current vertex and therefore remains inside the corresponding facet-star union.

Whenever T occurs, the induction hypothesis places the current support in the union of the two facet stars adjacent to the base edge. Mochán's Corollary 5.2 then says that the current facet still has e as its base edge. Hence T exchanges u with v and exchanges w with q , preserving the asserted regions.

This proves the claim for r_i , Tr_iT and Tr_p . It also covers arbitrary interleavings. For $p < n-1$, T commutes with r_p , so the inverse of Tr_p has the same form. At the boundary $p = n-1$ one must also check

$$(Tr_{n-1})^{-1} = r_{n-1}T.$$

Here r_{n-1} first changes the support inside the facet star of the current endpoint; Corollary 5.2 then applies before T acts. Thus the inverse preserves the same invariant. $\square$

The localization lemma distinguishes which endpoint a voltage can reach once the voltage is also a monodromy.

Lemma 4 (endpoint classification). *Let $h \in H_P$.*

1. *If h is a monodromy of M_n , then the first-coordinate vertex of Φh is u .*
2. *If yh is a monodromy of M_n , then the first-coordinate vertex of Φyh is v .*

Proof. Suppose first that h is a monodromy and that Φh has first-coordinate vertex v . Monodromies commute with the automorphism ρ_1 , so

$$\Phi^1 h = (\Phi h) \rho_1.$$

The automorphism ρ_1 fixes u, w and exchanges v, q on the base square. Thus the right-hand side has first-coordinate vertex q , contradicting Lemma 3, which places the vertex of $\Phi^1 h$ in $\{u, v\}$.

For the second assertion, the construction gives

$$\Phi y = \Phi, \quad \Phi^1 y = \Phi^{010}. \quad (7)$$

If yh were a monodromy taking the first-coordinate vertex of Φ to u , commutation with ρ_1 would force $\Phi^1 yh$ to have first-coordinate vertex u . But by (7) and Lemma 3 the latter vertex lies in $\{w, q\}$. Hence the first-coordinate vertex of Φyh must be v . □

6 The missing $k = 1$ intersection

Proposition 5. *Let $n \geq 4$ and let X be the normalized two-vertex premaniplex (2). For every $0 \leq m \leq n$ and every $s, t \in \{a, b\}$,*

$$\xi(\Pi_{[0,m]}^{s,t}(X)) \cap \xi(\Pi_{[1,n]}^{s,t}(X)) = \xi(\Pi_{[1,m]}^{s,t}(X)). \quad (8)$$

Proof. The case $m = n$ is tautological. Assume $m < n$ and begin at $s = a$. Let ω belong to the left-hand side of (8). Since ω is the voltage of a lower path using colors in $[0, m]$, formula (5) identifies it with a monodromy in $\langle r_0, \dots, r_m \rangle$. In particular, it fixes all faces of ranks greater than m .

An upper voltage from a to a has the form $h \in H_P$, whereas one from a to b has the form yh with $h \in H_P$. By Lemma 4, if

$$\Phi \omega = (\psi, x),$$

then its first-coordinate vertex is u in the closed case and v in the open case.

We next upgrade this first-coordinate statement to equality of the complete rank-0 face in $M_n = \widehat{2}M_{n-1}$. If $m < n - 1$, the lower monodromy fixes the facet, hence $x = 0$. Suppose $m = n - 1$ and abbreviate

$$U = \mathcal{F}(u), \quad V = \mathcal{F}(v), \quad W = \mathcal{F}(w), \quad Q = \mathcal{F}(q).$$

In the closed case, direct localization gives

$$\text{supp}(x) \subseteq U \cup V.$$

Apply the same localization to $\Phi^1 \omega = (\Phi \omega) \rho_1$ and pull the support back by ρ_1 . This gives

$$\text{supp}(x) \subseteq U \cup Q.$$

Every facet containing both v and q contains their join, namely the base two-face, and hence also contains u . Therefore

$$(U \cup V) \cap (U \cup Q) = U.$$

Thus $\text{supp}(x) \subseteq U$, which means $(u, x) = (u, 0)$ as rank-0 faces of the $\widehat{2}$ construction.

In the open case, use (7). The direct test gives support in $U \cup V$, while the test from $\Phi^1 y = \Phi^{010}$, followed by the ρ_1 pullback, gives support in $V \cup W$. Every facet containing u and w contains the base two-face and hence v , so

$$(U \cup V) \cap (V \cup W) = V.$$

Consequently $(v, x) = (v, 0)$ as rank-0 faces.

In the closed case, Φ and $\Phi\omega$ now have the same rank-0 face and the same faces of all ranks greater than m . Strong flag-connectivity of the intervening section supplies a flag path R from Φ to $\Phi\omega$ using only colors $[1, m]$. Let g be its monodromy word and let $W = p(R)$. Then W is an a -to- a path and (5) gives $\xi(W) = g^{-1}$. Since $\Phi g = \Phi\omega$ and M_n is regular, $g = \omega$; hence the reversed projected path W^{-1} has voltage ω .

In the open case, $\Phi^0 = \Phi r_0$ and $\Phi\omega$ have the same rank-0 face and the same faces above rank m . Let R be the resulting $[1, m]$ path from Φ^0 to $\Phi\omega$, with monodromy word g . Its projection $W = p(R)$ runs from b to a , and (5) gives $\xi(W) = g^{-1}r_0$. Therefore W^{-1} runs from a to b and has voltage r_0g . The equality $\Phi r_0g = \Phi\omega$, together with regularity of M_n , implies $r_0g = \omega$. Thus W^{-1} is the required path. If $P \cap [1, m]$ is empty, no $[1, m]$ path can switch the two vertices of X . The supposed intersection element would therefore produce an impossible projected path, so both sides of (8) are empty in this boundary case.

This proves (8) for paths starting at a . The color-preserving automorphism σ of X that interchanges a and b satisfies, for every dart d ,

$$\xi(\sigma d) = r_0 \xi(d) r_0. \quad (9)$$

For the top color this follows because T commutes with r_0 . Conjugating the three voltage sets in (8) by r_0 proves the two remaining ordered endpoint cases. The reverse inclusion in (8) is automatic, completing the proof. $\square$

Corollary 6. *For every normalized core (2), both ξ and the augmented assignment ξ' satisfy the full intersection condition (1).*

Proof. Proposition 5 proves $k = 1$. The case $k = 0$ and the case $m = n$ are tautological. Every case $k > 1$ is [3, Theorem 6.4]. This proves (1) for ξ .

If $m < n$, neither a lower path nor the reconstructed $[k, m]$ path uses the top color, so both act trivially on the auxiliary cyclic coordinate of ξ' . The projection argument of [3, Corollary 6.5] therefore lifts (1) from ξ to ξ' . The case $m = n$ remains tautological. $\square$

By [2, Theorem 4.2], the derived maniplex $X^{\xi'}$ is polytopal. The finite cyclic augmentation, [4, Propositions 19 and 20] and [4, Theorems 29 and 31] ensure, for a sufficiently large finite parameter, that no extra automorphism merges the two fibers. Hence its full symmetry type graph is exactly X .

7 The rank-four base cases

When the distance between the extreme link colors is $n = 3$, the only normalized cases not already covered by one or two links or by an interval are

$$L(X) = \{0, 1, 3\}, \quad L(X) = \{0, 2, 3\}. \quad (10)$$

We verified these cases using the regular toroidal polyhedron $\{4, 4\}_{(8,0)}$, with 512 flags, as the rank-three input. The separating involutions are represented by the color words

$$2101210212 \quad \text{and} \quad 210212,$$

respectively. For both types the verification checks:

1. the defining distant-color square relations and the original base-flag conditions;
2. that every semi-edge voltage is a nonidentity involution;
3. that parallel link colors have distinct voltages at each endpoint;
4. every nontrivial pair $k = 1, 2, 3$ and $m = 0, 1, 2$ in (1);
5. all four ordered endpoint pairs;
6. the known two-link type $\{0, 3\}$ as a positive control;
7. failure after replacing the top voltage by the identity as a destructive negative control.

Two implementations reconstruct the flags, reflections, base flags and voltage darts independently. They do not import one another. For the largest upper group, of order 2^{21} , both use SymPy's deterministic Schreier–Sims implementation for membership testing; their group-theoretic kernels are therefore not independent. Every explicit group closure is preceded by an exact order calculation and refused above 4096; every remaining reachable-state enumeration has the same 4096-item hard cap. The supplied runner executes the programs serially, monitors the complete process tree, and terminates it if sampled private memory exceeds 1024 MiB or the run exceeds 120 seconds.

The resource guard itself has a destructive test: a 1 MiB limit terminates the process tree and leaves no Python process behind. Under the standard guard, complete reproduction used 64.8 MiB and 58.1 MiB of peak private memory for the two implementations, with running times 10.47 seconds and 4.90 seconds. Both exited with code zero, every exact intersection comparison passed, and the negative control failed as required.

This section is a computer-assisted finite lemma. It does not replace the general proof in Sections 4–6, and it depends on the correctness of the two implementations and of their shared SymPy membership kernel.

8 Proof of the main theorem

Proof. Let X be a connected rank- r two-vertex premaniplex and let $L = L(X)$ be its nonempty link-color set.

All rank-three types are known to be realized by finite abstract polyhedra. If $|L| \leq 2$, or if L is an interval, the claim is [3, Theorem 6.9]. Otherwise let $i = \min L$ and $j = \max L$. Restrict to the colors $[i, j]$ and shift their labels so that the extreme links become 0 and $n = j - i$.

If $n \leq 2$, the core is among the already known cases. If $n = 3$, the only remaining cores are (10), which are polytopal by Section 7. If $n \geq 4$, Corollary 6 and the voltage criterion produce a finite abstract polytope whose full symmetry type graph is the normalized core.

Every color below i or above j is necessarily a semi-edge color. Repeated application of the standard $\widehat{2}$ rank-raising operation and its dual restores these exterior semi-edge colors. These operations preserve finiteness and the number of flag orbits and add precisely the required semi-edge to the symmetry type graph [3, Section 4]; [4, Proposition 11]. Thus the resulting finite abstract polytope has full symmetry type graph X . $\square$

9 Scope and provenance

The contribution claimed here is Theorem 1, specifically the arbitrary-internal-link localization in Lemmas 2–4 and the resulting $k = 1$ intersection proof in Proposition 5. The voltage criterion, the recursive regular polytopes M_n , the assignments ξ and ξ' , the $k > 1$ intersection theorem, and the automorphism-suppression augmentation are cited ingredients from [2, 3, 4].

Theorem 1 strengthens the maniplex existence theorem of [4] to abstract polytopes and completes the two-vertex cases left open after [3]. The structural theory of two-orbit polytopes developed by Hubard and Schulte [5] is complementary: it studies the internal structure and automorphism groups of two-orbit polytopes, rather than supplying this realization theorem.

The result concerns symmetry type graphs with exactly two vertices. It does not classify the polytopes realizing a given type, bound their minimum size, or address arbitrary symmetry type graphs with three or more vertices.

References

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AI-assisted research and writing disclosure

Carptopus is the author responsible for the mathematical claims and the public version of this manuscript. Codex was used as an AI-assisted research, verification and writing tool.